

Smoothing topology optimization results using pre-built lookup tables

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Abstract

Topology optimization techniques are typically performed on a design domain discretized with finite element meshes to generate efficient and innovative structural designs. The optimized structural topologies usually exhibit zig-zag boundaries formed from straight element edges. Existing techniques to obtain smooth structural topologies are limited. Most methods are computationally expensive, as they are performed iteratively with topology optimization. Other methods, such as post-processing methods, are applied after topology optimization, but they cannot guarantee obtaining equivalent structural designs, as the volume and geometric features may be changed. This study presents a new method that uses pre-built lookup tables to transform the shape of boundary elements obtained from topology optimization to create smoothed structural topologies. The new method is developed based on the combination of the bi-directional evolutionary structural optimization (BESO) technique and marching geometries to determine structural topologies and lookup tables, respectively. An additional step is used to ensure that the generated result meets a target volume. A variety of 2D and 3D examples are presented to demonstrate the effectiveness of the new method. This research shows that the new method is highly efficient, as it can be directly added to the last step of topology optimization with a low computational cost, and the volume and geometric features can be preserved in smoothed topologies. Finite element models are also created for original and smoothed structural topologies to show that the structural stiffness can be significantly enhanced after smoothing.

Keywords: Topology optimization, Smoothing, Finite element analysis, Volume-preserving, Area-preserving

1. Introduction

Topology optimization is a structural optimization technique to obtain innovative structures that possess lightweight, high-performance and cost-efficient material characteristics [1]. As such, topology optimization techniques have been widely adopted across many disciplines, including additive manufacturing [2, 3, 4], architectural applications [5, 6, 7], furniture designs [8, 9], and biological materials [10, 11].

Topology optimization techniques typically use finite element analysis (FEA) to evaluate structural performance. To this end, many topology optimization techniques require the continuous design domain to be discretized into finite element meshes in order to conduct FEA, referring

to element-based topology optimization approaches [12, 13]. Popular element-based approaches include the solid isotropic material with penalization (SIMP) method [14, 15] and the bi-directional evolutionary structural optimization (BESO) method [16, 17]. The key difference between these two methods is the density representation of their elements. The SIMP method allows the densities of elements to vary continuously between 0 and 1 [18], while the BESO method only allows the densities of elements to be either 0 or 1 [16]. As a result, optimal structural topologies obtained from the SIMP method may possess “grey” boundaries corresponding to “soft” elements; BESO structural topologies may possess zig-zag boundaries formed from straight element edges [19]. It is worth pointing out that the resolution of these element-based topologies can be significantly increased using a large number of elements, which can create finer

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grey and zig-zag boundaries, and lead to visually smooth boundaries.

To generate smooth structural topologies, grey and zig-zag boundaries typically require further modifications [20]. Two dominant strategies in recent studies are *geometrically constrained topology optimization* methods and *post-processing* methods. The former strategy employs geometric operations during the optimization process, where smooth structural topologies are generated in each iteration [19, 21, 22, 23, 24, 25, 26]. However, such methods are computationally expensive due to the generation of unnecessary smoothed topologies throughout the optimization [20]. The latter strategy focuses on smoothing the generated optimal topologies, such as iso-surface extraction algorithms [27, 28, 29, 30] and Laplacian smoothing techniques [31, 32]. However, such methods cannot create smoothed models that are truly equivalent to the actual topology optimization results, as the volume or geometric features cannot be preserved in the generated models [20]. Baccialia et al. [33] have proposed a volume-preserved smoothing method that can obtain equivalent smooth models, but their method results in a performance loss of about 2%. Recently, Li et al. [34] have used a volume-preserved smoothing method to obtain smooth structural topologies. However, their method has not been extended to commonly used irregular elements such as tetrahedra.

For practical applications, computer-aided design (CAD) models are typically needed. However, the smoothed topologies fail to be directly converted into CAD models, especially in 3D cases [35]. They are recognized as non-manifold models in CAD software because their volumetric meshes include coincident faces and singularities [33, 35]. The most straightforward strategy to overcome this problem is to rebuild a smoothed manifold CAD model [35]. In doing so, the generated model may differ from the actual topology optimization result regarding the volume and structural performance.

This paper presents a novel post-processing method (or smoothing method) to create smoothed manifold CAD models from element-based topology optimization results. The proposed method can transform the shape of elements using pre-built lookup tables straight after the topology optimization is completed, which can effectively and conveniently smooth the zig-zag boundaries. The new method is developed considering the equivalency of the original and smoothed structural topolo-

gies, meaning that the geometric features and volume can be preserved after smoothing. Section 2 first briefly summarizes the method of topology optimization and describes the details of the new smoothing method. Section 3 demonstrates the implementation of the new method and compares the differences between original and smoothed structural topologies. Section 4 discusses the effect of different mesh types and sizes in potential practical applications, followed by a conclusion in Section 5.

2. Methodology

2.1. BESO structural topology optimization

This study uses the soft-kill BESO method to perform structural topology optimization [16, 36]. For a discretized design domain with N elements, the compliance minimization (or stiffness maximization) problem statement subject to a volume constraint can be written as follows.

$$\text{Minimize : } C = \frac{1}{2} \mathbf{U}^T \mathbf{K} \mathbf{U} \quad (1a)$$

$$\text{Subject to : } \mathbf{V}^* - \sum_{i=1}^N \mathbf{V}_i x_i = 0 \quad (1b)$$

$$x_i = x_{min} \text{ or } 1 \quad (1c)$$

where C is the compliance of the structure, $\mathbf{K}$ is the global stiffness matrix, $\mathbf{U}$ is the displacement vector, $\mathbf{V}^*$ is the target structural volume, and $\mathbf{V}_i$ is the volume of the i -th element. The design variable, x_i , determines whether the i -th element is solid ($x_i = 1$) or void ($x_i = x_{min} = 0.001$).

The material model of elements is defined as a function of the element density.

$$E(x_i)^k = x_i^p E_0 \quad (2)$$

where k is the current iteration number, $E(x_i)^k$ is the Young's modulus of i -th element, E_0 is the design Young's modulus of elements, and $p = 3$ is the penalty exponent.

To relate the structural performance of elements to the optimization objective, i -th elemental sensitivity number, $S_e(i)$, can be defined as

$$S_e(i) = -\frac{1}{p} \frac{\partial C}{\partial x_i} = \begin{cases} \frac{1}{2} \mathbf{u}_i^T \mathbf{K}_i^0 \mathbf{u}_i & \text{when } x_i = 1 \\ \frac{x_{min}^{p-1}}{2} \mathbf{u}_i^T \mathbf{K}_i^0 \mathbf{u}_i & \text{when } x_i = x_{min} \end{cases} \quad (3)$$

where $\mathbf{K}_i^0$ is the elemental stiffness matrix, and $\mathbf{u}_i$ is the displacement vector.

To improve the convergence of the BESO technique, historical sensitivity numbers can be averaged, where

$$\hat{S}_e(i) = \frac{S_e^k(i) + S_e^{k-1}(i)}{2} \quad (4)$$

The target volume for the next iteration, $\mathbf{V}^{k+1}$, depends on the current volume, $\mathbf{V}^k$, and a specified evolutionary ratio, ert , where

$$\mathbf{V}^{k+1} = \mathbf{V}^k(1 \pm ert) \quad (5)$$

Note that a threshold sensitivity number, S_{th} , is calculated based on $\mathbf{V}^{k+1}$ using the method described in [36]. Elements that possess the sensitivity number higher and lower than S_{th} are then preserved and removed, respectively. The optimization process is repeated until reaching the target volume and satisfying the convergence criteria [16].

2.2. Smoothing preparation

Once the element-based structural topology is obtained using the method described in Section 2.1, the nodal sensitivity field, S_n , is calculated for smoothing preparation. This can be achieved by employing a filter on elements [37, 38], as shown in Figure 1, where the filter covers a circular domain Ω_p with a center of node p . The filtering scheme is given as

$$S_n(p) = \frac{\sum_{q=1}^{N_n} (r_{min} - r_{pq}) \hat{S}_e(q)}{\sum_{q=1}^{N_n} (r_{min} - r_{pq})} \quad (6)$$

where $S_n(p)$ is the sensitivity number of node p . N_n is the number of element centers covered by the filter. r_{min} is the filter radius, and r_{pq} is the distance between the center of element q and node p . It is noted that filtering all elements gives a nodal sensitivity field, S_n .

Finally, an iso-value, S_{iso} , is specified between the maximum and minimum S_n , that is, within the nodal sensitivity field, for further smoothing procedures as described below.

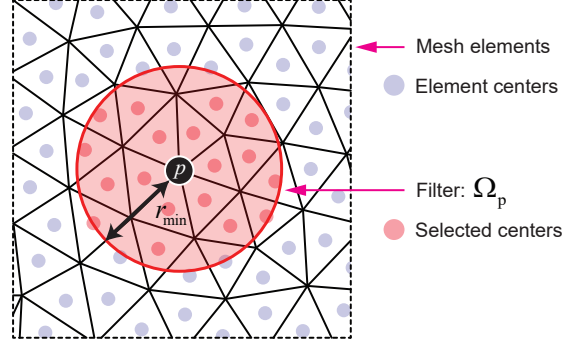


Figure 1: Obtaining the nodal sensitivity field by employing a filter on mesh elements. This figure shows that the filter covers a circular domain Ω_p with a center of node p , which has a radius of r_{min} .

2.3. Smoothing 2D structural topologies

This sub-section describes a method to directly transform the shape of elements generated from topology optimization, so that smooth 2D structural topologies can be conveniently created. For a 2D structural topology, its nodal sensitivity field forms a level-set function [25, 39]. Specifying S_{iso} determines the solid domain Ω_S , void domain Ω_V and boundary Γ , as shown in Figure 2(c). For node p in the design domain, its (x, y, z) position, $\mathbf{p}$, related to S_{iso} can be described as

$$\begin{cases} \mathbf{p} \in \Omega_V, & \text{if } S_n(p) > S_{iso} \\ \mathbf{p} \in \Gamma, & \text{if } S_n(p) = S_{iso} \\ \mathbf{p} \in \Omega_S, & \text{if } S_n(p) < S_{iso} \end{cases} \quad (7)$$

$St(p)$ is introduced to determine whether $S_n(p)$ corresponds to solid, void, or boundary using 0 and 1 only, where

$$St(p) = \begin{cases} 0, & \text{if } S_n(p) > S_{iso} \\ 1, & \text{if } S_n(p) \leq S_{iso} \end{cases} \quad (8)$$

This equation represents that void points are those with $St(p) = 0$, and both solid and boundary points possess $St(p) = 1$.

To transform the shape of structural elements, this study uses lookup tables to pre-define all possible transformation results of a unit element. For quadrangular elements shown in Figure 2(a), the corresponding lookup table can be built by modifying the Marching Squares (MS) algorithm [40, 41]. More specifically, the iso-lines in the

original marching squares are replaced with pre-defined triangular or quadrangular meshes to represent solid domains, as shown in Figure 2(b). Using Equation 8, elemental nodes can be assigned with $St(p) = 1$ or 0 for internal (red) and external (grey) points, respectively. Additional boundary (orange) points are then created on the midpoint between internal and external points. Mathematically, there are 16 possible transformation results of a quadrangular element, calculated using

$$Cs(i) = \sum_{j=0}^{N_i} St(n_j)2^j \quad (9)$$

where Cs is the case number, N_i is the number of nodes of element i , and $St(n_j)$ represents the 0/1 state of j th node in element i ; the number of possible cases is calculated as $2^{N_i} = 2^4 = 16$. However, there are actually 18 marching squares listed in the lookup table, as shown in Figure 2(b).

It should be noted that the calculated Cs value may correspond to more than one marching geometry [42]. Such special cases occur when the diagonal corners are determined as internal (red) points, with the intermediate region surrounded by boundary (orange) points being either filled or vacant, resulting in connected and disconnected solid domains, respectively. In the present example, special cases are $Cs = 5$ and $Cs = 10$, with each containing two possible marching squares. Therefore, the total number of possible marching squares is $16 + 2 = 18$. In this study, the selection of marching squares for special cases can be simply determined based on the averaged elemental sensitivity, $\hat{S}_e(i)$ (see Equation 4). Specifically, $\hat{S}_e(i) \leq S_{iso}$ and $\hat{S}_e(i) > S_{iso}$ correspond to connected and disconnected solid domains, respectively.

Next, all elements in the design domain are reshaped using the modified MS lookup table. Finally, for each boundary (orange) point t between an internal (red) point A and an external (grey) point B , its (x, y, z) coordinate is relocated to a new position $\mathbf{t}_{new}$ using the following linear interpolation method, as shown in Figure 2.

$$\mathbf{t}_{new} = \mathbf{A} + \frac{S_{iso} - S_n(A)}{S_n(B) - S_n(A)}(\mathbf{B} - \mathbf{A}) \quad (10)$$

where $\mathbf{A}$ and $\mathbf{B}$ are the (x, y, z) locations of the internal point A and external point B , respectively. $S_n(A)$ and $S_n(B)$ are filtered nodal sensitivity numbers of points A and B , respectively. It is noted

that relocating all boundary points gives a smooth boundary.

Furthermore, the intersections between the generated structural topology and the design domain are unchanged after smoothing. This is because the elements located on the boundaries of the design domain possess $St = 1$ on their nodes.

To summarize, Equation 6 creates a nodal sensitivity field, S_n , for a 2D structural topology. The specified iso-value, S_{iso} , determines the condition of each node, where $St = 0$ or 1, as described in Equations 8. In doing so, elements can be reshaped using the modified MS lookup table based on Equation 9. Then, the shapes of the transformed elements are further updated using linear interpolation, as shown in Equation 10. This 2D smoothing is demonstrated more clearly in Figure 2.

2.4. Smoothing 3D structural topologies

3D structural topologies can also be smoothed based on different lookup tables, depending on the mesh element type. The two most common lookup tables are Marching Cubes (MC) [43] and Marching Tetrahedra (MT) [44] for hexahedral and tetrahedral mesh elements, respectively. Compared with the original MC and MT methods, the 3D transformation cases are pre-defined by closed surface meshes to represent solid domains, as shown in Figure 3(a) and (b). It is worth pointing out that the 8-nodes hexahedral element can be transformed into 260 possible configurations, but Figure 3(a) only shows 17 unique cases; the remaining $260 - 17 = 243$ cases can be obtained by rotating and mirroring these 17 cases. Among these 17 unique cases, $Cs = 65$ is the special case, where it contains two possible marching cubes, highlighted in Figure 3(a). Similar to the previous 2D example, connected and disconnected solid domains for the present 3D case are selected when $\hat{S}_e(i) \leq S_{iso}$ and $\hat{S}_e(i) > S_{iso}$, respectively.

The entire smoothing process for 3D structural topologies is similar to the workflow developed for 2D cases. First, S_n is calculated for all nodes using Equation 6. Second, S_{iso} is specified to assign all nodes with $St = 0$ or 1 based on S_n , as described in 8. Third, all elements are reshaped from the lookup table according to St using Equation 9. Fourth, the shape of transformed elements are further modified using Equation 10. Different to 2D smoothing, an additional step is required in 3D cases.

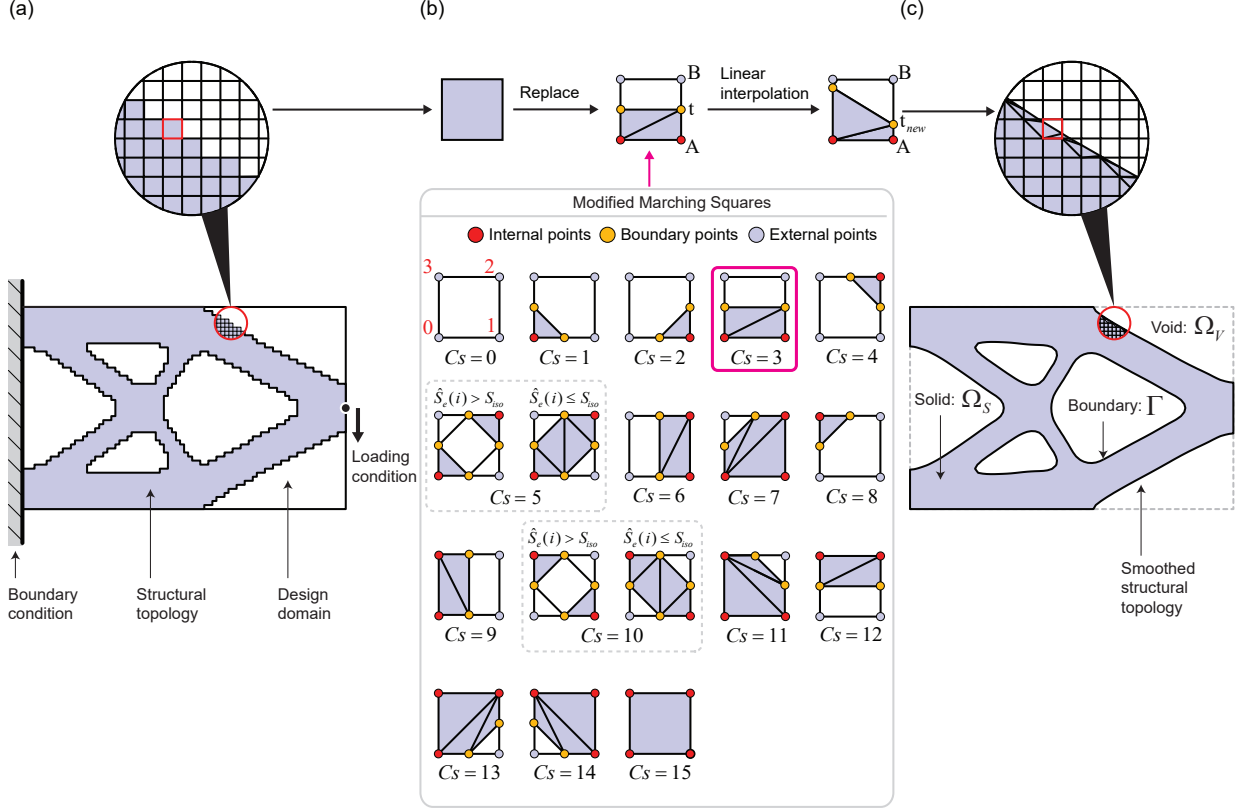


Figure 2: Smoothing the structural topology of a 2D cantilever. (a) Initial setup of the optimization problem, where the design domain is discretized with quadrangular elements. (b) The lookup table is built using the modified Marching Squares algorithm that has 18 possible configurations. (c) The transformation result is obtained through reshaping all original elements.

As shown in Figure 4, this study removes all internal faces of 3D structural topologies. It should be highlighted that all transformed elements are closed surface meshes. They form a smoothed configuration that includes coincident faces, as shown in Figure 4(a). These coincident faces are identified as internal faces. Removing all internal faces gives 3D surface meshes, as shown in Figure 4(b), which can be considered as CAD-compatible manifold meshes.

2.5. Surface remeshing

The products of the above 2D and 3D smoothing processes are surface meshes [45]. As shown in Figure 5, the quality of these surfaces can be improved through “surface remeshing”. It should be noticed that many remeshing techniques are developed based on manifold CAD models like surface meshes [46].

This study uses the incremental remeshing algorithm described in [45, 47, 48]. This algorithm can be found in many CAD software, such as Rhino 7. The remeshing algorithm repeatedly modifies the mesh edges and vertices based on local operations until all edge lengths are close to the prescribed target value; the local operations include splitting long edges, collapsing short edges, flipping mesh edges, and relocating vertices. As a result (see Figure 5), the elements on the surface can be re-distributed uniformly using triangular elements toward the target edge length. It can also be seen that the geometric features, such as corners and sharp edges, are preserved during the remeshing process.

2.6. Preserving area or volume

The smoothed configurations obtained from the above methods can be understood as iso-sensitivity models, as they are determined based on the S_{iso}

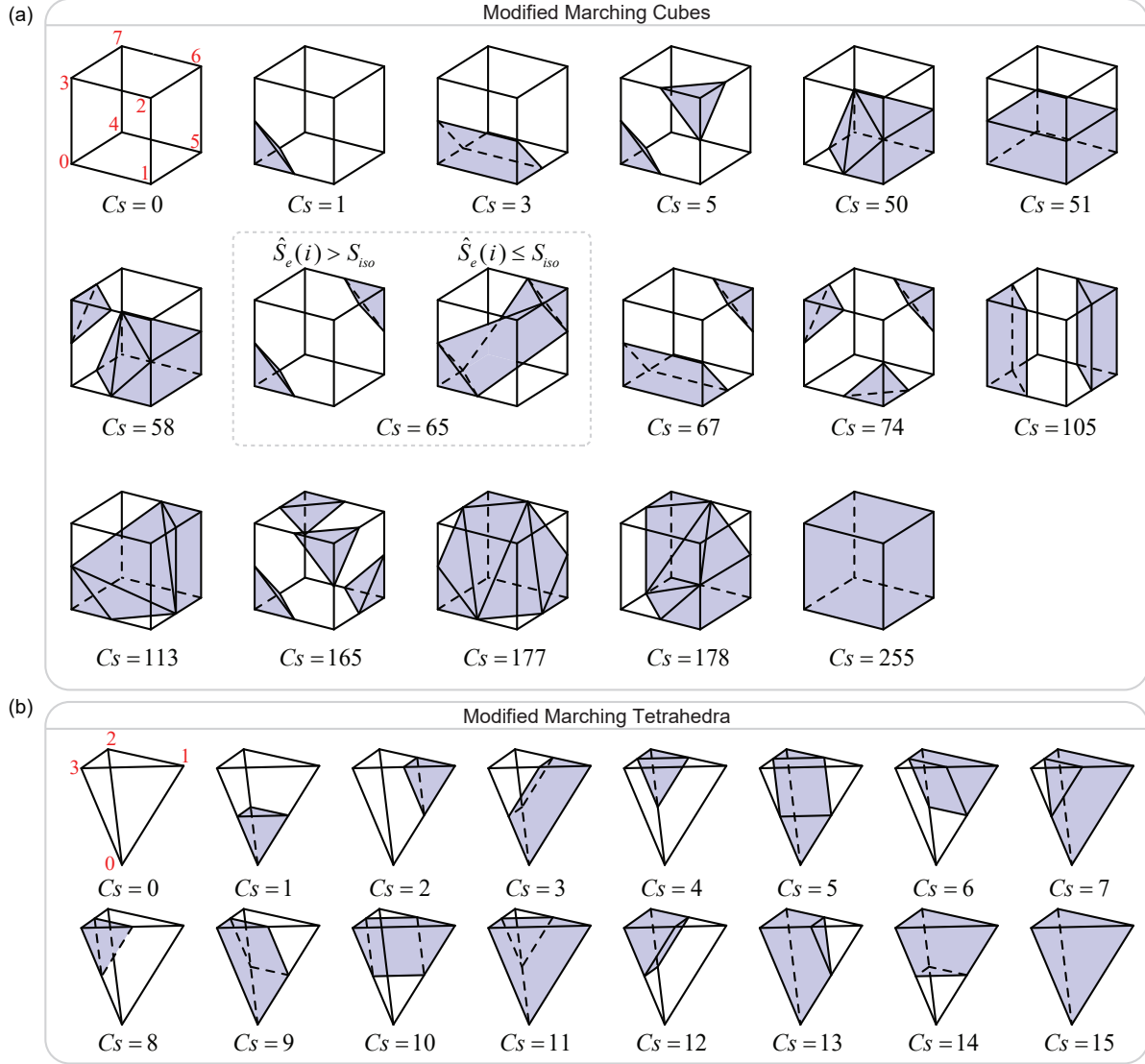


Figure 3: Modified 3D lookup tables. The red numbers represent the indices of the elemental nodes. (a) 17 unique cases in the modified Marching Cubes method of a transformed hexahedral element. (b) All 16 possible cases in the modified MT method of a transformed tetrahedral element.

value. Moreover, changing S_{iso} can directly influence the final area and volume of 2D and 3D iso-sensitivity models, respectively. In order to meet a target area or volume, this study treats S_{iso} as a variable and introduces the bisection method to modify S_{iso} automatically.

$$S_{iso} = \frac{S_{up} + S_{lo}}{2} \quad (11)$$

where $S_{lo} = 0$ and $S_{up} = 2 \times S_{th}$ are the initial lower and upper bound, respectively. The S_{th} is the threshold sensitivity number mentioned in Section 2.1. The bounds are updated using

$$\begin{cases} S_{lo} = S_{iso}, & \text{if } V > V^* \\ S_{up} = S_{iso}, & \text{if } V \leq V^* \end{cases} \quad (12)$$

where V is the current volume/area of the iso-sensitivity model, and V^* is the target volume/area.

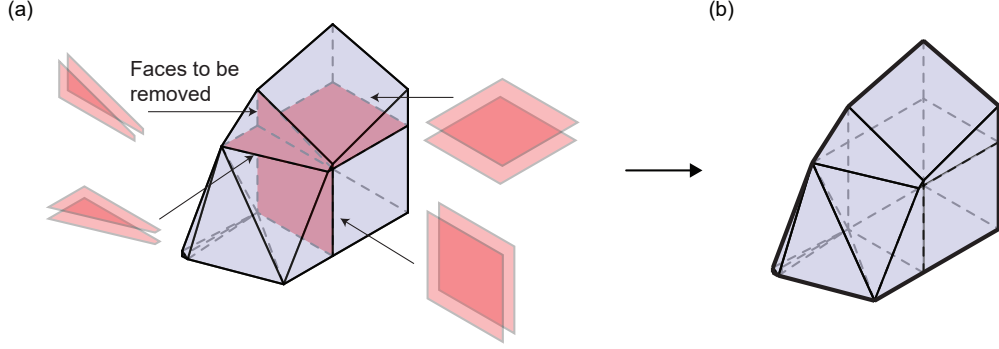


Figure 4: Removing coincident faces of 3D elements. (a) All coincident faces are identified as internal faces, which are required to be removed. (b) Removing all internal faces gives a 3D closed surface mesh.

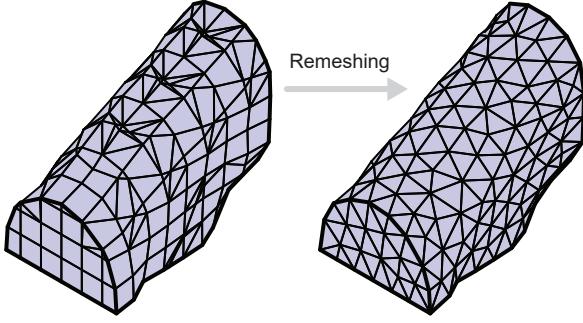


Figure 5: Improve the surface quality using the incremental remeshing algorithm so that the elements on the surface can be re-distributed uniformly based on a target edge length and preserve the geometric features.

The bi-sectional iterative process is repeated until

$$\frac{|V - V^*|}{V_D} \leq V_{tol} \quad (13)$$

where V_D is the total volume/area of the design domain, $V_{tol} = 0.1\%$ is the allowable tolerance. This equation simply represents that the final structural topology will possess the total volume/area close to the target value with an acceptable difference. The entire smoothing process that considers preserving area or volume process is summarized in the flowchart shown in Figure 6.

3. Numerical analysis

A numerical analysis method is developed to examine the capabilities of the above smoothing

methods. More specifically, the obtained BESO structural topologies are compared with their corresponding smoothed results to observe any variations in shape and structural performance.

This study uses the commercial finite element analysis (FEA) software, Abaqus, and a Python script to perform structural analysis and BESO topology optimization, respectively, with details fully described in [36]. Once the optimal structural topologies are obtained, a C# code is used to execute the above algorithms for generating smoothed models in the Rhino 7 CAD software. The original and smoothed optimal structural topologies are then imported into Abaqus to perform FE structural analysis to compare their stiffness performance. In order to have a fair comparison, their FE meshes are set to have the same mesh type and target edge length. Specifically, 2D and 3D cases are analyzed using triangular and tetrahedral elements, respectively, with the target edge length set to be the same value as used in the remeshing process. Note that equivalent topologies are discretized using the same meshing algorithm embedded in the commercial Abaqus software to create FE models; meshing algorithms of FE models are not included in the scope of this study.

The material used in this study is assumed to be isotropic and linear elastic, with Young's modulus of $E = 1$ MPa and Poisson's ratio of $\nu = 0.3$. Unless otherwise stated, BESO topology optimization parameters are: $ert = 2\%$, $p = 3$, and $r_{min} = 3$ mm.

The 3D smoothing process is first tested using a cantilever example, as shown in Figure 7. With reference to Figure 7(a), the design domain of the

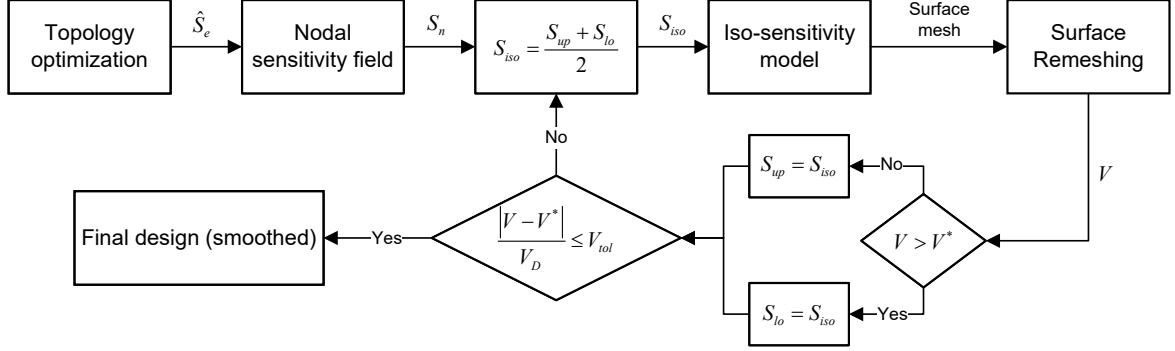


Figure 6: The flowchart of the entire smoothing process. The process of preserving area or volume will ensure that the smoothed result meets a target area or volume.

cantilever is $60 \text{ mm} \times 10 \text{ mm} \times 40 \text{ mm}$. A $F = -1 \text{ N}$ point load is applied at the center of the free end. A fixed boundary condition is assigned behind the whole cantilever. The objective volume fraction of BESO is set as 15%. Smoothed configurations are obtained from structural topologies with different mesh sizes, as shown in Figures 7(a)-(b) and (c)-(d). For original BESO structural topologies, solid cubic elements that have an edge length of 1 mm and 0.5 mm are used, as shown in Figures 7(a) and (c), respectively. For smoothed structural topologies, original elements are reshaped using the modified MC lookup table and then the assemblies are remeshed to a surface mesh with triangular elements that have an average edge length of 0.5 mm, as shown in Figures 7(b) and (d). All structural topologies are discretized using solid tetrahedral elements with an edge length of 0.5 mm in FEA, so that the compliance values, C , are comparable.

In Figure 7, it can be seen that the zig-zag boundaries of original structural topologies have been successfully smoothed. It can also be seen that the geometrical features have been preserved, where the shapes of the original and smoothed structural topologies are almost identical. Moreover, the structural volumes are changed within an allowable tolerance. Together, it can be concluded that the proposed smoothing method can generate equivalent structural topologies to the original topology optimization results.

Furthermore, the compliance of Figures 7(a)-(d) are 5.336 Nmm, 4.855 Nmm, 5.060 Nmm, and 4.973 Nmm, respectively. It is seen that the smoothed

structures are stiffer, corresponding to lower compliance. More specifically, the compliance of Figures 7(a) and (c) are reduced by 9.02% and 1.73% after smoothing, respectively. The enhanced structural performance may be attributed to the stress being more uniformly distributed in the smoothed models, thus resulting in lower compliance, demonstrated more clearly in Figures 7(a)-(b). However, the improvement is less noticeable using a finer mesh, as the resolution of the original structural topology is already high, meaning that the shape is similar to its corresponding smoothed form, as shown in Figures 7(c)-(d). Nevertheless, this example still demonstrates that an enhanced structural performance can be obtained through smoothing.

4. Discussion

4.1. Regular and irregular 2D meshes

This subsection examines the structural performance of smoothed topologies obtained from regular and irregular 2D quadrangular meshes [49], as shown in Figures 8(a) and (b), respectively.

Figure 8(a) shows a $240 \text{ mm} \times 40 \text{ mm}$ MBB beam initially meshed with 1 mm square elements. A $F = -1 \text{ N}$ point load is applied at the midpoint of the top edge. For topology optimization, the objective volume fraction is set as 50%, and r_{min} is changed to 2 mm. During the smoothing process, the modified MS lookup table is used to obtain transformed quadrangular elements, and then they

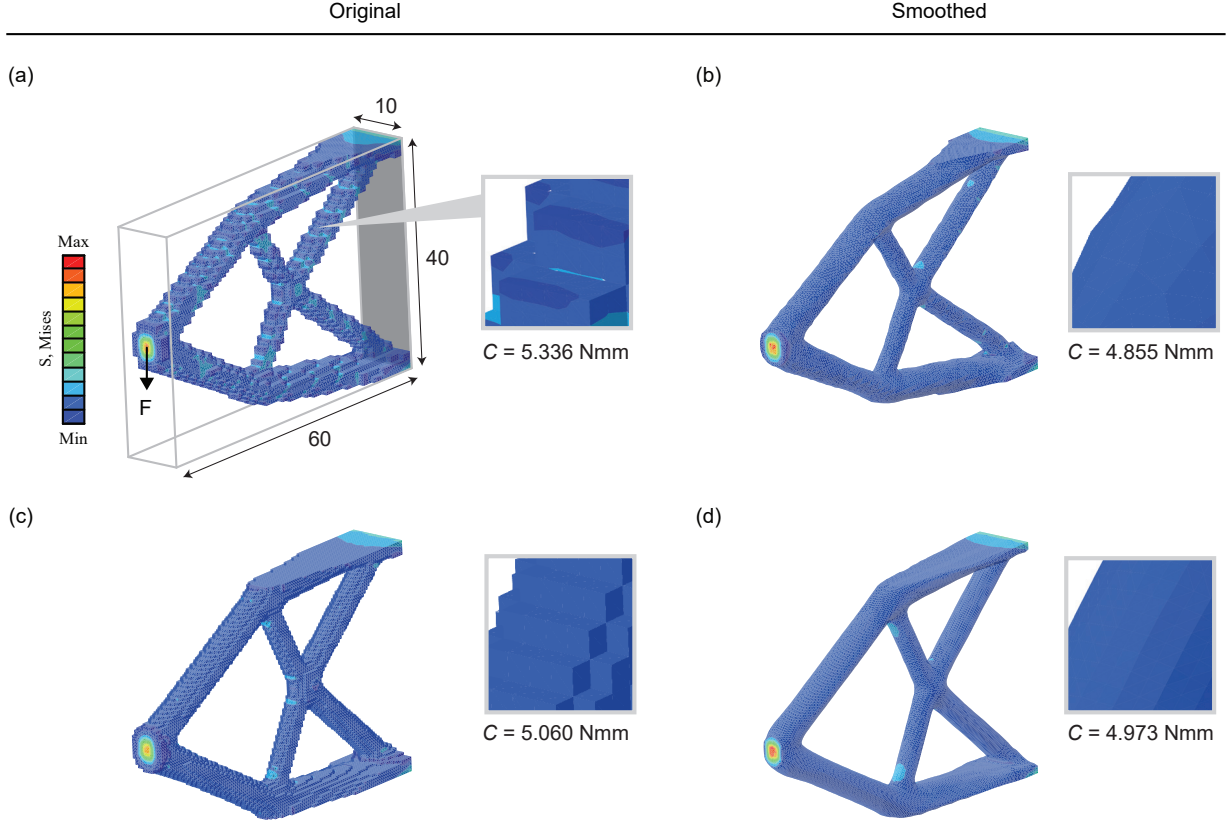


Figure 7: FE results of original and smoothed structural topologies. (a)-(b) shows that cubic elements that have an edge length of 1 mm are smoothed into a surface mesh with triangular elements that have an average edge length of 0.5 mm. (c)-(d) shows that cubic elements that have an edge length of 0.5 mm are smoothed into a surface mesh with triangular elements that have an average edge length of 0.5 mm. (a) and (c) are original structural topologies; (b) and (d) are smoothed structural topologies. In FEA (a)-(d) are meshed using solid tetrahedral elements with an average edge length of 0.5 mm.

are remeshed using triangular elements with an average edge length of 1 mm. For FE structural analysis, the original and smoothed structural topologies are both discretized using triangular elements with an average edge length of 1mm.

In Figure 8(a), it can be seen that the MBB beam has been successfully smoothed using the proposed 2D smoothing method, which reduces the original compliance by 3.46% (from 94.465 Nmm to 89.998 Nmm). The same example has been smoothed previously using the FTOP method [23] with a final compliance of 89.68 Nmm, which is similar to the present smoothed result in shape and compliance. However, the FTOP method requires more than 300 iterations to obtain the final smoothed result with smoothed structural topologies generated in each iteration. In contrast, the proposed smoothing

method is much more efficient, which can be added to the last step of the BESO process. This example only uses 70 iterations to obtain the structural topology, and the total time spent on the smoothing process is only 0.43 seconds using an ordinary laptop with an Intel i7-9750H 2.60GHz processor.

The 777 mm $\times$ 713 mm chair example shown in Figure 8(b) is initially meshed with irregular quadrangular elements in order to create smooth boundaries, resulting in all elements having different shapes with an averaged edge length of 4.25 mm. The chair is loaded at its non-design domains with a relative force density of $F : F' = 3 : 2$ [50], where $F = 0.03$ N and $F' = 0.02$ N. The objective volume fraction is set as 33% to obtain the BESO structural topology. In terms of smoothing process, the modified MS lookup table in Figure 2(b) is used,

where each sub-figure may be reshaped depending on the shape of the selected irregular element using Equation 10. Finally, the transformed elements are remeshed using triangular elements that have an average edge length of 3.5 mm. For FE structural analysis, the original and smoothed structural topologies are both discretized using triangular elements with an average edge length of 3.5 mm.

In Figure 8(b), it is seen that the chair has been successfully smoothed, meaning that the proposed smoothing method is capable of dealing with irregular meshes. Similar to the above examples with regular meshes, the compliance also reduces after smoothing, where a 2.93% improvement is achieved; C is reduced from 88.729 Nmm to 86.133 Nmm. It is worth pointing out that complex geometries are typically discretized with irregular meshes to perform topology optimization. Using the proposed method, optimal structural topologies of such complex geometries can be conveniently smoothed, which means that the proposed method is highly beneficial for complex geometries initially discretized with irregular meshes.

4.2. 3D practical application

A 771 mm $\times$ 400 mm $\times$ 713 mm 3D chair example is given here to demonstrate the potential practical applications of the proposed smoothing method, as shown in Figure 9. Figure 9(a) shows the setup of the optimization problem, from which 81% of the material is removed through topology optimization. It is assumed that the chair carries an 80 kg adult, hence the non-design domain is loaded with $F = 0.0003$ N/mm² and $F' = 0.0002$ N/mm² on the seat part and the back-rest part, respectively [50]. Four 20 mm $\times$ 20 mm regions on the bottom surface are assigned to have a fixed boundary condition. r_{min} is changed to 30 mm to obtain the BESO structural topology. The modified MT lookup table is used for smoothing. The original and smoothed chairs are both meshed with solid tetrahedra that have an edge length of 10 mm.

Figures 9(b)-(c) show the original and smoothed structural topologies, respectively. It is seen that the smoothed result improves not only the visual effect but also the structural performance, where the compliance of the original result is reduced by 6.52% (from 61.231 Nmm to 57.235 Nmm). Following the discussion of Section 4.1, complex 3D geometries are typically discretized with irregular

tetrahedral meshes. After performing topology optimization on such complex 3D geometries, the proposed smoothing method can then be employed to create manifold CAD models that possess smooth boundaries, as shown in Figures 9(c)-(d). Therefore, it is confirmed that the proposed smoothing method can be effectively adopted for 3D applications without the limitation of the mesh element type.

5. Conclusion

In this work, a new post-processing method is proposed, which uses pre-built lookup tables to smooth the zig-zag boundaries obtained from topology optimization. It is shown that the modified Marching Squares (MS), Marching Cubes (MC), and Marching Tetrahedra (MT) provide effective lookup tables for structures initially discretized with quadrangular (2D), hexahedral (3D), and tetrahedral (3D) elements, respectively. A series of examples are tested to show that: (1) the new method can create smooth structural topologies that preserve the volume and geometric features from the original optimal topologies, and (2) the compliance can be reduced after smoothing to achieve stiffer structures. Moreover, the new method is shown to be highly efficient, as it is performed as a post-processing step after the topology optimization is completed. This study also shows that the proposed method supports various mesh element types. Finally, a complex 3D example is used to demonstrate that the new method can be readily applied to a wide range of practical applications.

CRediT authorship contribution statement

Zhi Li: Conceptualization, Methodology, Software, Data Curation, Validation, Visualization, Writing - Original Draft. **Ting-Wei Lee:** Writing - Original Draft, Visualization, Writing - Review & Editing. **Yuan Yao:** Conceptualization, Methodology. **Yi Min Xie:** Conceptualization, Methodology, Supervision, Writing - Review & Editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relation-

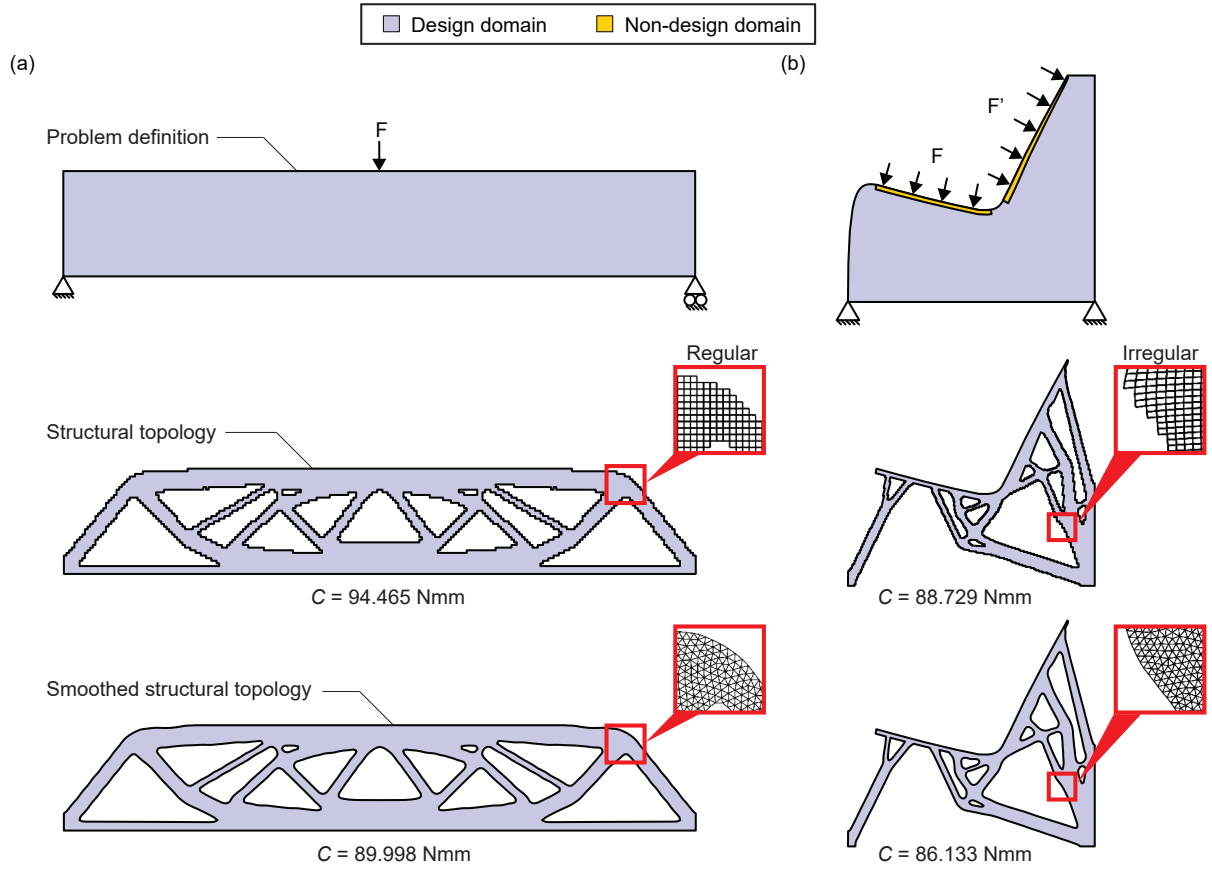


Figure 8: Smoothing 2D structural topologies with regular and irregular meshes. Sub-figures from the top are: optimization problem, original structural topology, and smoothed structural topology. (a) MBB beam example initially meshed with square elements. (b) Chair example initially meshed with irregular quadrangular elements.

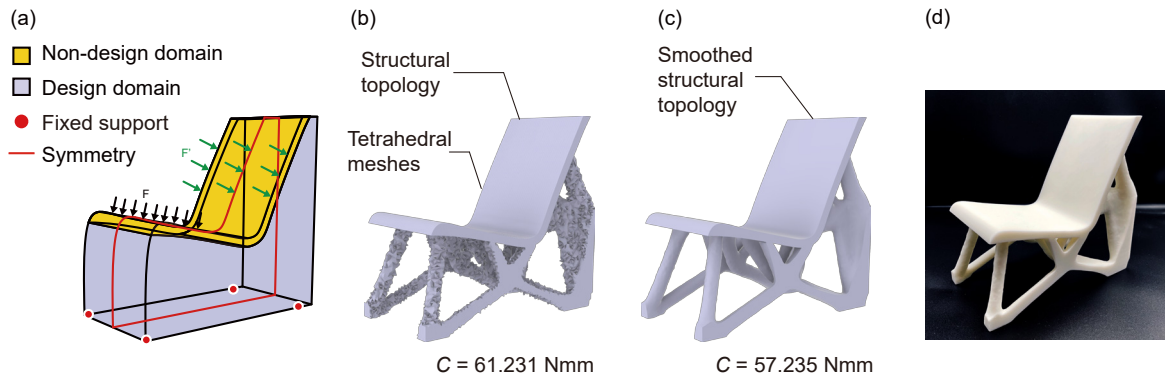


Figure 9: Potential practical applications of the proposed smoothing method demonstrated with a 3D chair example. (a) Initial setup of the optimization problem. (b) Original structural topology obtained from topology optimization. (c) Smoothed structural topology. (d) Manufactured 3D model.

ships that could have appeared to influence the work reported in this paper.

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